

$$\begin{bmatrix} 1 & 0 & 0 & -7 \\ -1 & 1 & 0 & 5 \\ 2 & -5 & 1 & 2 \end{bmatrix} \xrightarrow[\text{+ Row 1}]{\text{Row 2 = Row 2}} \begin{bmatrix} 1 & 0 & 0 & -7 \\ 0 & 1 & 0 & -2 \\ 2 & -5 & 1 & 2 \end{bmatrix} \xrightarrow[\text{- 2 \times Row 1}]{\text{Row 3 = Row 3}} \begin{bmatrix} 1 & 0 & 0 & -7 \\ 0 & 1 & 0 & -2 \\ 0 & -5 & 1 & 16 \end{bmatrix}$$

$$\xrightarrow[\text{+ 5 \times Row 2}]{\text{Row 3 = Row 3}} \begin{bmatrix} 1 & 0 & 0 & -7 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 6 \end{bmatrix} \quad \therefore y = \begin{bmatrix} -7 \\ -2 \\ 6 \end{bmatrix}$$

$$[U \ y] : \begin{bmatrix} 3 & -7 & -2 & -7 \\ 0 & -2 & -1 & -2 \\ 0 & 0 & -1 & 6 \end{bmatrix} \xrightarrow[\text{\times (-1)}]{\text{Row 3 = Row 3}} \begin{bmatrix} 3 & -7 & -2 & -7 \\ 0 & -2 & -1 & -2 \\ 0 & 0 & 1 & -6 \end{bmatrix} \xrightarrow[\text{+ 2 \times Row 3}]{\text{Row 1 = Row 1}} \begin{bmatrix} 3 & -7 & 0 & -19 \\ 0 & -2 & -1 & -2 \\ 0 & 0 & 1 & -6 \end{bmatrix}$$

$$\begin{bmatrix} 3 & -7 & 0 & -19 \\ 0 & -2 & -1 & -2 \\ 0 & 0 & 1 & -6 \end{bmatrix} \xrightarrow[\text{+ Row 3}]{\text{Row 2 = Row 2}} \begin{bmatrix} 3 & -7 & 0 & -19 \\ 0 & -2 & 0 & -8 \\ 0 & 0 & 1 & -6 \end{bmatrix} \xrightarrow[\text{\times -1/2}]{\text{Row 2 = Row 2}} \begin{bmatrix} 3 & -7 & 0 & -19 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & -6 \end{bmatrix}$$

$$\begin{bmatrix} 3 & -7 & 0 & -19 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & -6 \end{bmatrix} \xrightarrow[\text{+ 7 \times Row 2}]{\text{Row 1 = Row 1}} \begin{bmatrix} 3 & 0 & 0 & 9 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & -6 \end{bmatrix} \xrightarrow[\text{\times 1/3}]{\text{Row 1 = Row 1}} \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & -6 \end{bmatrix}$$

$$\therefore x = \begin{bmatrix} 3 \\ 4 \\ -6 \end{bmatrix}$$

Row Reduction Method

By Theorem 3 from Chapter 1, the matrix equation $Ax=b$ has the same solution set as the system of equations corresponding to the augmented matrix $[A \ b]$.

$$\begin{bmatrix} 3 & -7 & -2 & -7 \\ -3 & 5 & 1 & 5 \\ 6 & -4 & 0 & 2 \end{bmatrix} \xrightarrow[\text{+ Row 1}]{\text{Row 2 = Row 2}} \begin{bmatrix} 3 & -7 & -2 & -7 \\ 0 & -2 & -1 & -2 \\ 6 & -4 & 0 & 2 \end{bmatrix} \xrightarrow[\text{- 2 \times Row 1}]{\text{Row 3 = Row 3}} \begin{bmatrix} 3 & -7 & -2 & -7 \\ 0 & -2 & -1 & -2 \\ 0 & 0 & 4 & 16 \end{bmatrix}$$

$$\begin{bmatrix} 3 & -7 & -2 & -7 \\ 0 & -2 & -1 & -2 \\ 0 & 0 & 4 & 16 \end{bmatrix} \xrightarrow[\text{+ 5 \times Row 2}]{\text{Row 3 = Row 3}} \begin{bmatrix} 3 & -7 & -2 & -7 \\ 0 & -2 & -1 & -2 \\ 0 & 0 & -1 & 6 \end{bmatrix} \xrightarrow[\text{\times (-1)}]{\text{Row 3 = Row 3}} \begin{bmatrix} 3 & -7 & -2 & -7 \\ 0 & -2 & -1 & -2 \\ 0 & 0 & 1 & -6 \end{bmatrix}$$

$$\begin{bmatrix} 3 & -7 & -2 & -7 \\ 0 & -2 & -1 & -2 \\ 0 & 0 & 1 & -6 \end{bmatrix} \xrightarrow[\text{+ 2 \times Row 3}]{\text{Row 1 = Row 1}} \begin{bmatrix} 3 & -7 & 0 & -19 \\ 0 & -2 & -1 & -2 \\ 0 & 0 & 1 & -6 \end{bmatrix} \xrightarrow[\text{+ Row 3}]{\text{Row 2 = Row 2}} \begin{bmatrix} 3 & -7 & 0 & -19 \\ 0 & -2 & 0 & -8 \\ 0 & 0 & 1 & -6 \end{bmatrix}$$